Tianchi Luan

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Education

Beihang University — M.Sc. in Computer Science 2024–2027 (expected)
State Key Laboratory of Virtual Reality Technology and Systems
Thesis: 3D Scene Reconstruction Methods Integrating Large Multimodal Models.
Supervisor: Yue Qi.

Beijing Jiaotong University — B.Sc. in Computer Science 2020–2024

Research Interests

3D Scene Reconstruction and Understanding; Visual Geometry Models; Extended Reality (XR).

Research Experience

ParS3R: Hierarchical Semantic 3D Scene Reconstruction

First author; manuscript under review at AAAI 2027.

- Developed a feed-forward 3D reconstruction framework with hierarchical decoding, combining a frozen geometry backbone with trainable appearance features to recover object-to-part semantics from unposed multi-view RGB images.
- Designed and implemented 3D cross-view hierarchy consolidation and tree-distance-aware prototypical triplet learning for fine-grained part discrimination and cross-view consistency.
- Implemented the training and evaluation pipelines and conducted cross-dataset comparisons and ablations on PartScene, a benchmark covering 50 scenes from ScanNet++, Replica, and Tanks and Temples.

Feed-Forward Geometry for Mobile AR Tour Deployment

First author; SSRN preprint.

- Built a Unity/AR Foundation mobile client and Flask backend, integrating pretrained IGGT predictions for 3D semantic mapping, object-level annotation, and AR tour deployment.
- Implemented confidence-weighted $\mathrm{Sim}(3)$ alignment for metric scale recovery, together with reference-view maintenance based on geometric confidence and co-visibility for map-aligned camera pose refinement.
- Evaluated localization on 7-Scenes and Cambridge Landmarks and conducted a 20-participant study with AR-experienced users. Compared with the manual workflow, mean deployment time decreased from 6.3 to 2.3 minutes indoors and from 9.5 to 2.7 minutes outdoors.

Publications

Bold: applicant; * equal contribution; † corresponding author.

Preprints

[1] **T. Luan**, Y. Wang, J. Wang, C. Wang†, M. Liu, and Y. Qi. *Feed-Forward Visual Geometry Model Meets AR: An On-the-fly Mobile AR Tour Deployment Framework with 3D Semantic Mapping and Camera Pose Refinement*. Preprint. SSRN. 2026. DOI: 10.2139/ssrn.7619744.

Under Review

[2] **T. Luan***, Y. Wang*, and Y. Qi†. "ParS3R: Feed-Forward Part-Aware Hierarchical Semantic 3D Scene Reconstruction". Under review at AAAI 2027. 2026.

Honors and Awards

Graduate Academic Scholarships, Beihang University
Second Class (Year 1); Third Class (Years 2–3).

Second-Class Academic Scholarship, Beijing Jiaotong University
Undergraduate Year 1.

Technical Skills

Programming: C, C++, C#, Python.

3D and AR: Unity, AR Foundation, Blender.

Backend Development: Django, Flask.

Design and Documentation: Adobe Illustrator, Photoshop, Premiere, PowerPoint, LaTeX.

Languages: Chinese (native), English.

Certification: CCF Certified Software Professional (CSP), top 6.29% in the certification session (2023).

Teaching and Academic Service

Teaching Assistant, Visual Modeling Methods 2024–2025; 2025–2026

Reviewer, ICASSP 2027